Chapter 9. Comparing Tests

Power, most powerful tests, Neyman–Pearson, nonparametric tests and their efficiency

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9.1 Power and the choice of a test

Among level-α tests of $H_0 : \theta \in \Omega_0$ against $H_1 : \theta \in \Omega_1$, a test ϕ^* is **uniformly most powerful** (UMP) if $E_\theta \phi^* \geq E_\theta \phi$ for every $\theta \in \Omega_1$ and every level-α test ϕ.

9.2 Most powerful tests

Theorem 9.1 (Neyman–Pearson lemma)

For simple hypotheses $H_0 : \theta = \theta_0$ and $H_1 : \theta = \theta_1$ with joint densities f_0, f_1, the test

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & f_1(\mathbf{x}) > k f_0(\mathbf{x}), \\ \gamma, & f_1(\mathbf{x}) = k f_0(\mathbf{x}), \\ 0, & f_1(\mathbf{x}) < k f_0(\mathbf{x}), \end{cases}$$

with $k \geq 0$ and $\gamma \in [0, 1]$ chosen so that $E_0 \phi^* = \alpha$, is most powerful at level α.

Proof. For any level-α test ϕ, $(\phi^* - \phi)(f_1 - k f_0) \geq 0$ pointwise: where $f_1 > k f_0$, $\phi^* = 1 \geq \phi$; where $f_1 < k f_0$, $\phi^* = 0 \leq \phi$. Integrating, $E_1 \phi^* - E_1 \phi \geq k(E_0 \phi^* - E_0 \phi) = k(\alpha - E_0 \phi) \geq 0$. ■

So the most powerful test rejects for large likelihood ratios f_1 / f_0. Randomization at the boundary is needed only for discrete data.

Example 9.1

$N(\mu, 1)$, $H_0 : \mu = 0$ against $\mu = 1$: $f_1 / f_0 = \exp(n\bar{x} - n/2)$ is increasing in $\bar{x}$, so the MP test rejects when $\bar{X} > z_\alpha / \sqrt{n}$. It does not depend on the value 1, so it is UMP against $\mu > 0$.

Theorem 9.2 (One-parameter exponential families)

If $f(x; \theta) = \exp\{g(\theta) T(x) - B(\theta) + S(x)\}$ with g increasing, the test that rejects when $\sum T(x_i) > c$ (randomizing at $= c$), with size α at θ_0, is UMP for $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$.

Proof. For any $\theta_1 > \theta_0$, the ratio $f(\mathbf{x}; \theta_1) / f(\mathbf{x}; \theta_0)$ is increasing in $\sum T$. So ϕ^* is the MP test of θ_0 against θ_1, for every $\theta_1 > \theta_0$. Its power is nondecreasing in θ: for $\theta' < \theta''$, ϕ^* is MP of size $\alpha' = E_{\theta'} \phi^*$ for θ' against θ'', hence at least as powerful as the trivial test $\phi \equiv \alpha'$. Therefore $\sup_{\theta \leq \theta_0} E_\theta \phi^* = \alpha$, and enlarging H_0 to $\theta \leq \theta_0$ keeps the level. ■

The same holds whenever the family has a **monotone likelihood ratio** in a statistic T. For two-sided alternatives no UMP test exists in general; one restricts to unbiased tests (power $\geq \alpha$ on H_1), and the equal-tail tests are then UMP unbiased in exponential families.

9.3 Large-sample power and efficiency of tests

Theorem 9.3

Let a level-α test of $H_0 : \theta = \theta_0$ against $\theta > \theta_0$ reject when $\sqrt{n}\{T_n - \mu(\theta_0)\} / \sigma(\theta_0) \geq z_\alpha$, where

$\sqrt{n}\{T_n - \mu(\theta)\}/\sigma(\theta) \xrightarrow{d} N(0,1)$ locally uniformly. For local alternatives $\theta_n = \theta_0 + K/\sqrt{n}$,

$$\gamma_n(\theta_n) \to 1 - \Phi\Big(z_\alpha - K\frac{\dot\mu(\theta_0)}{\sigma(\theta_0)}\Big),$$

and the sample size needed for power $1-\beta$ at θ_1 is approximately

$$N \approx \Big(\frac{\sigma(\theta_0)}{\dot\mu(\theta_0)}\Big)^2\Big(\frac{z_\alpha + z_\beta}{\theta_1 - \theta_0}\Big)^2.$$

Proof. Under θ_n, $\sqrt{n}\{T_n - \mu(\theta_0)\}/\sigma(\theta_0) = \sqrt{n}\{T_n - \mu(\theta_n)\}/\sigma(\theta_0) + \sqrt{n}\{\mu(\theta_n) - \mu(\theta_0)\}/\sigma(\theta_0)$. The first term is approximately $N(0,1)$, since $\sigma(\theta_n) \to \sigma(\theta_0)$. The second tends to $K\dot\mu(\theta_0)/\sigma(\theta_0)$, which gives the power limit. For the sample size, set $1 - \Phi(z_\alpha - \sqrt{N}(\theta_1-\theta_0)\dot\mu/\sigma) = 1-\beta$, i.e. $z_\alpha - \sqrt{N}(\theta_1-\theta_0)\dot\mu/\sigma = -z_\beta$, and solve for N. ■

The quantity $e = (\dot\mu(\theta_0)/\sigma(\theta_0))^2$ is the **efficacy** of the test. The **Pitman ARE** of test 1 relative to test 2 is the limiting inverse ratio of sample sizes, $N_2/N_1 \to e_1/e_2$.

9.4 Nonparametric tests and their efficiency

In the location model $f(x-\theta)$ with f symmetric about 0, consider $H_0: \theta = 0$ against $\theta > 0$.

- **Sign test:** $S = \sum I(X_i > 0) \sim B(n, \tfrac12)$ under H_0 for every continuous f (distribution-free). With $T_n = S/n$, $\mu(\theta) = 1 - F(-\theta)$, $\dot\mu(0) = f(0)$ and $\sigma(0) = \tfrac12$, so the efficacy is $4f(0)^2$.
- **Wilcoxon signed-rank test:** $W^+ = \sum_i R_i^+ I(X_i > 0)$, where R_i^+ is the rank of $|X_i|$. Under H_0 its distribution is that of $\sum_k k B_k$ with B_k i.i.d. Bernoulli($\tfrac12$), with mean $\frac{n(n+1)}{4}$ and variance $\frac{n(n+1)(2n+1)}{24}$. Its efficacy is $12(\int f^2)^2$.
- **t-test** (or z-test): efficacy $1/\sigma^2$.

Proposition 9.4

$ARE(\text{sign}, t) = 4\sigma^2 f(0)^2$, $ARE(W^+, t) = 12\sigma^2(\int f^2)^2$. For the normal these are $2/\pi \approx 0.64$ and $3/\pi \approx 0.955$; for the Laplace, 2 and 1.5; and $ARE(W^+, t) \geq 0.864$ for every symmetric f.

Proof. Take ratios of efficacies (Theorem 9.3). For the signed-rank statistic, $W^+/\binom{n+1}{2}$ is asymptotically equivalent to the U-statistic estimating $P(X_1 + X_2 > 0)$. Its derivative at $\theta = 0$ is $2\int f^2$, and its null standard deviation is $1/\sqrt{3n}$, which gives efficacy $12(\int f^2)^2$. The normal and Laplace values follow by direct substitution. The lower bound $0.864 = 108/125$ is attained by a parabolic density (Hodges–Lehmann). ■

So distribution-free tests lose little under normality and can gain a lot under heavy tails. The two-sample analogues (Wilcoxon rank-sum versus two-sample t) have the same AREs.